HONG ZIYANG

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SUMMARY

Motivated Applied AI graduate with a strong foundation in AI, data analysis and problem solving, seeking opportunities to apply technical skills to impactful challenges. Proactive and adaptable, with a continuous growth mindset and a willingness to take initiative in dynamic environments.

EDUCATION

Singapore Institute of Technology ***Aug 2023 - Apr 2026***

Bachelor of Science (with Honours) in **Applied Artificial Intelligence** ***Singapore***

*Degree Classification: **Honours with Distinction** (Conferred on 28 May 2026) | Grade: 4.18 / 5.00*

TECHNICAL SKILLS

Programming Languages: Python, R, SQL, HTML5/CSS, JavaScript

Platforms/Tools/Libraries: PyTorch, Scikit-learn, NumPy, Pandas, Hugging Face, TensorBoard, RStudio, Tidyverse, Shiny, Plotly, Streamlit, Gradio, Quarto, pgAdmin 4, MongoDB (NoSQL), MariaDB, Jupyter Notebook, Google Colab, VS Code, GitHub, Excel/PowerPoint/Word

SOFT SKILLS

Adaptability to Change, Growth Mindset, Problem Solving, Agility, Team Player, Leadership, Proactive and Effective Communication

WORK EXPERIENCE

AETHER with RAiD, RSAF ***Sep 2025 - Apr 2026***

AI Solution Intern, MARL Team ***Singapore***

- Restored arming, flight stability and episode resets in an inherited drone simulation environment, by debugging across ROS2, PX4, Unreal Engine and the reinforcement learning codebase.
- Enabled the transition from a single-agent setup to a multi-agent testbed, by implementing segmentation mask support for every drone and reworking the reward structure after the switch from LiDAR to depth camera.
- Improved reset reliability and the agents' learning signal, by fixing a PX4 and ROS2 race condition, correcting the state representation to use drone-to-target distance, and adding evaluation metrics to compare training runs.

TradeFlow Capital Management ***Jan 2025 - Jul 2025***

Project Member, Integrative Team Project ***Singapore***

- Engineered crop health features for yield prediction of selected agricultural commodities, by computing the Normalized Difference Vegetation Index (NDVI) from Sentinel-2 multispectral satellite imagery.
- Enabled models to learn cyclical crop growth seasons, by applying temporal encoding to calendar month.
- Added environmental context to the NDVI time series, by integrating temperature and precipitation data from a weather API.

The Republic of Singapore Air Force ***Feb 2022 - May 2023***

Service Medic cum Dispensary Department In-Charge, National Service ***Singapore***

- Led a team of 4 personnel to deliver efficient medical support, by optimising task delegation and coordination.
- Maintained optimal stock levels and minimised shortages, by managing inventory transactions and supply chain monitoring in SAP.

ASMPT Ltd ***Oct 2017 - Mar 2018***

Quality Assurance Auditor, Intern ***Singapore***

- Identified process deviations in production through quality assurance audits.
- Contributed to a reduction in defect rates and improved production efficiency, by addressing the identified deviations.

KEY PROJECTS

Hybrid RRT-MARL Framework for Autonomous Drone Swarm Navigation | *Capstone* ***Sep 2025 - Mar 2026***

- Benchmarked 4 approaches for 3-drone swarm navigation, with QMIX achieving 1.2% to 3.1% higher average reward and about 3.8x faster evaluation than MADDPG and MAPPO, and 6.9% higher reward than the supervised baseline, by implementing MAPPO, MADDPG, QMIX and a GRU-based supervised baseline in PyTorch under identical conditions.
- Enabled explicit information exchange between drones for MAPPO and the supervised baseline, by designing an attention-based communication module that encodes each drone's 12-dimensional observation into a 64-dimensional message, with training tracked in TensorBoard.

Multimodal LLM-Based Profanity Detection and Detoxification ***Jan 2025 - Mar 2025***

- Led a team of 5 to deliver a multimodal content moderation system for text and speech, by defining the system design and coordinating development through to deployment.
- Detected and rephrased profanity while preserving conversational context, by combining a RoBERTa classifier for context-aware detection with a T5 model for detoxification.
- Enabled real-time moderation of audio input, by integrating OpenAI Whisper and Microsoft SpeechT5 and deploying on Hugging Face Spaces with a Gradio interface.

FoodVision: YOLOv8 Food Detection ***Oct 2024 - Dec 2024***

- Achieved about 0.80 mAP50 (0.80 precision, 0.75 recall) detecting 55 food classes, by training a YOLOv8n model as part of a team of 4.
- Delivered calorie estimation per 100g of detected food, by building a Streamlit web app with image upload and camera capture, deployed on Hugging Face Spaces.

Speech Emotion Recognition using Wav2Vec2 ***Sep 2024 - Oct 2024***

- Achieved 69.02% accuracy classifying 4 emotions (neutral, happy, sad, angry) on the IEMOCAP dataset, by fine-tuning Wav2Vec2.
- Improved model robustness across different speakers, by applying data augmentation to the training audio.

Visualising the World's Biggest Data Breaches and Hacks Over Time ***Jun 2024 - Jul 2024***

- Led a team of 6 to deliver an interactive web application analysing global data breaches over time, improving on an existing bubble plot visualisation, by building it in R with Shiny and Plotly.
- Enabled user-driven trend analysis, by implementing filters by year and sector and switchable linear and logarithmic scales.

Augmented Reality (AR) Hazards Identification | *Final Year Project* ***May 2020 - Aug 2020***

- Led a team of 5 to deliver an AR hazard identification training prototype for an aviation industry partner, targeting hazard awareness in aircraft hangars, by developing in Unity with Vuforia SDK.
- Built the training content, by superimposing 3D hazard objects through QR code markers and adding interactive quizzes for knowledge evaluation.